FORMATIVE ASSESSMENT – 1 (FA-1)

MATHEMATICS PROJECT WORK

Topic: Number System & Properties of Rational Numbers

PRELIMINARY INFORMATION

Name of the Student: .....

Class: Class 8

Roll Number: .....

Name of the Lesson: Rational Numbers (Number System)

Name of the Guide Teacher: .....

AIM

To understand the classification of numbers in the number system and verify the commutative property of addition and multiplication for rational numbers.

OBJECTIVES

1. To identify Natural numbers, Whole numbers, Integers, and Rational numbers with clear examples.
2. To test the Commutative Property for addition and multiplication using simple rational numbers.
3. To represent a positive rational number accurately on a calibrated number line.

MATERIAL REQUIRED

- A4 white paper / Project ruled sheet
- Geometry box tools (Ruler scale, pencil, eraser)
- Blue and black writing pens

PROCEDURE

Step 1: Classification and Definitions of Numbers

1. Natural Numbers (N): Counting numbers starting from 1.
Examples: 1, 2, 3, 4, 5, ...

2. Whole Numbers (W): All natural numbers including zero (0).
Examples: 0, 1, 2, 3, 4, ...

3. Integers (Z): The collection of negative numbers, zero, and positive numbers.
Examples: ..., −3, −2, −1, 0, 1, 2, 3, ...

4. Rational Numbers (Q): Numbers written in the form $^{p}/_{q}$, where p and q are integers and q ≠ 0.
Examples: $^{1}/_{2}$, $^{3}/_{4}$, $-^{2}/_{5}$, 5, 0

PROCEDURE (CONTINUED)

Step 2: Verifying the Commutative Property

Let us take two simple rational numbers: **a = $^1/_2$** and **b = $^1/_4$**.

(i) Verification under Addition: [a + b = b + a]

LHS: a + b = $^1/_2 + {}^1/_4 = {}^{(2+1)}/_4 = {}^3/_4$

RHS: b + a = $^1/_4 + {}^1/_2 = {}^{(1+2)}/_4 = {}^3/_4$

✓ **Since LHS = RHS ($^3/_4 = {}^3/_4$), Addition is Commutative.**

(ii) Verification under Multiplication: [a × b = b × a]

LHS: a × b = $^1/_2 \times {}^1/_4 = {}^{(1\times1)}/_{(2\times4)} = {}^1/_8$

RHS: b × a = $^1/_4 \times {}^1/_2 = {}^{(1\times1)}/_{(4\times2)} = {}^1/_8$

✓ **Since LHS = RHS ($^1/_8 = {}^1/_8$), Multiplication is Commutative.**

Step 3: Number Line Representation of $^3/_4$

1. Draw a straight line using a ruler and pencil, marking arrows at both ends.
2. Mark reference points **0** and **1**.
3. Divide the unit length between 0 and 1 into **4 equal parts**.
4. Count 3 units to the right of 0 and plot the point $^3/_4$.



CONCLUSION

- The number system progresses naturally from Natural numbers to Whole numbers, Integers, and Rational numbers.
- The commutative property holds true for both addition and multiplication of rational numbers.
- Any rational number can be plotted on a number line by dividing the unit interval into equal sub-parts based on the denominator.

Signature of the Student: _____

Signature of the Guide Teacher: _____